

Ramnit Lab Walkthrough

Our intrusion detection system has alerted us to suspicious behavior on a workstation, pointing to a likely malware intrusion. A memory dump of this system has been taken for analysis. Your task is to analyze this dump, trace the malware's actions, and report key findings.

Q1. What is the name of the process responsible for the suspicious activity?

Use command `vol -f "file path name" ps.tree` also `Out-file ...\anything.txt` to see the output better in txt file.

```
"C:\Users\alex\Downloads\ChromeSetup.exe" C:\Users\alex\Downloads\ChromeSetup.exe
```

Suspicious process because it runs on user folder download.

Q2. What is the exact path of the executable for the malicious process?

```
"C:\Users\alex\Downloads\ChromeSetup.exe" C:\Users\alex\Downloads\ChromeSetup.exe
```

The path is this.

Q3. Identifying network connections is crucial for understanding the malware's communication strategy. What IP address did the malware attempt to connect to?

Use netscan command to find any network activity

TCPv4	192.168.19.133	49755	20.199.120.85	443	ESTABLISHED	3356	svchost.exe	2024-02-01
TCPv4	192.168.19.133	49697	95.100.200.202	443	CLOSE_WAIT	5912	WWAHost.exe	2024-02-01
TCPv4	192.168.19.133	49758	13.67.191.143	443	CLOSED 7312		MicrosoftEdgeU	2024-02-01
TCPv4	192.168.19.133	49691	192.229.221.95	80	CLOSE_WAIT	5912	WWAHost.exe	2024-02-01
TCPv4	192.168.19.133	49708	20.189.173.16	443	CLOSED 5912		WWAHost.exe	2024-02-01
TCPv4	192.168.19.133	49760	2.18.162.9	443	ESTABLISHED	2500	svchost.exe	2024-02-01
TCPv4	192.168.19.133	49699	95.100.200.202	443	CLOSE_WAIT	5912	WWAHost.exe	2024-02-01
TCPv4	192.168.19.133	49696	95.100.200.202	443	CLOSE_WAIT	5912	WWAHost.exe	2024-02-01
TCPv4	192.168.19.133	49682	58.64.204.181	5202	CLOSED 4628		ChromeSetup.ex	2024-02-01
TCPv4	192.168.19.133	49682	58.64.204.181	5202	SYN_SENT	4628	ChromeSetup.ex	2024-02-01

We look for the suspicious file activity and refer to the network connection

58.64.204.181

Q4. To determine the specific geographical origin of the attack, Which city is associated with the IP address the malware communicated with?

58.64.204.181 was not found in our database

This IP was reported **0** times. Confidence of Abuse is **0%**:

0%

ISP	NWT iDC Data Service
Usage Type	Data Center/Web Hosting/Transit
ASN	AS17444
Domain Name	hkbn.net
Country	 Hong Kong
City	Hong Kong

Refer to Abuseipdb answer is Hong Kong

Q5. Hashes serve as unique identifiers for files, assisting in the detection of similar threats across different machines. What is the SHA1 hash of the malware executable?

windows.filescan the malware

```
PS C:\Users\analyst\Desktop\Tools\volatility3-develop > python .\vol.py `
>> -f C:\Users\analyst\Desktop\temp_extract_dir\memory.dmp `
>> windows.filescan
>> Select-String -Pattern "ChromeSetup.exe"
Progress: 100.00 PDB scanning finished
0xca82b85307f0 \Users\alex\Downloads\ChromeSetup.exe
```

windows.dumpfiles --virtaddr <address>

```
FLARE-VN 07/22/2020 13:17:01
PS C:\Users\analyst\Desktop\Tools\volatility3-develop > python .\vol.py `
>> -o C:\Users\analyst\Desktop\temp_extract_dir\ChromeSetup_file `
>> -f C:\Users\analyst\Desktop\temp_extract_dir\memory.dmp `
>> windows.dumpfiles --virtaddr 0xca82b85307f0
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Cache FileObject FileName Result
DataSectionObject 0xca82b85307f0 ChromeSetup.exe Error dumping file
ImageSectionObject 0xca82b85307f0 ChromeSetup.exe file.0xca82b85307f0
```

Get-FileHash *ImageSectionObject* -Algorithm SHA1

```
PS C:\Users\analyst\Desktop\Tools\volatility3-develop > Get-ChildItem `
>> C:\Users\analyst\Desktop\temp_extract_dir\ChromeSetup_file `
>> -File |
>> Get-FileHash -Algorithm SHA1
```

Algorithm	Hash	Path
SHA1	280C9D36039F9432433893DEE6126D72B9112AD2	C:\Users\analyst\Desktop\temp_extract_dir\ChromeSetup_file\ChromeSetup_file.exe

Get the hash the .exe img one

280C9D36039F9432433893DEE6126D72B9112AD2

Q6. Examining the malware's development timeline can provide insights into its deployment. What is the compilation timestamp for the malware?

History ⓘ	
Creation Time	2019-12-01 08:36:04 UTC
First Submission	2024-02-03 00:02:57 UTC
Last Submission	2026-07-21 19:04:07 UTC
Last Analysis	2026-04-30 09:40:35 UTC

Submit the hash to VirusTotal and check details tab show creation time

2019-12-01 08:36

Q7. Identifying the domains associated with this malware is crucial for blocking future malicious communications and detecting any ongoing interactions with those domains within our network. Can you provide the domain connected to the malware?

Contacted Domains (4) ⓘ	
Domain	Detections
ddos.dnsnb8.net	15 / 91
dnsnb8.net	14 / 91
res.public.onecdn.static.microsoft	0 / 91
www.microsoft.com	0 / 91

Check the relation tab to show domain names associated with the malware.